UNIT 5

Applications, Use Cases and the Quantum Future

Unit 5 at a glance

Healthcare & drug discovery · materials science · logistics & optimization · quantum sensing & timing · industry case studies (IBM, Google, Microsoft, PsiQuantum) · ethics, society, policy · adoption challenges: cost, skills, standards · careers · education/research & India's opportunity

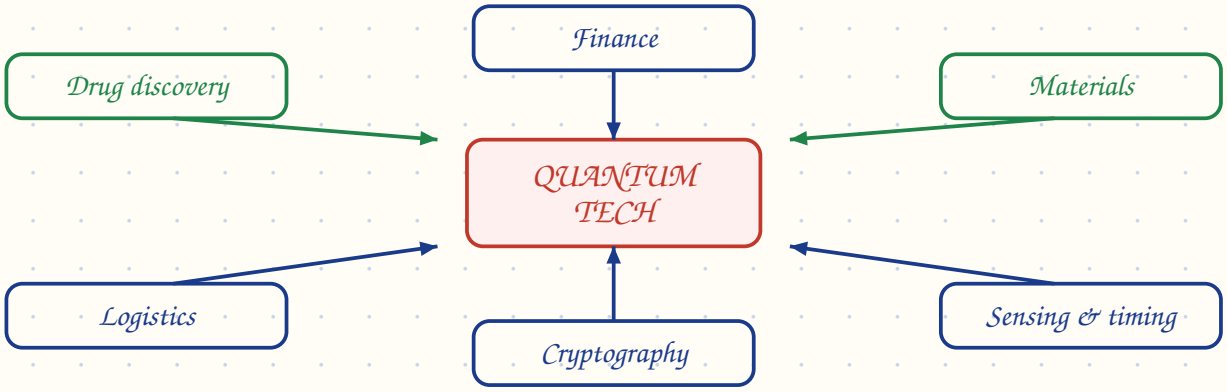


Fig 1: Where quantum technologies are expected to create value

1. Real-World Application Domains

Quantum technologies are not meant to replace classical computers. They target specific problems where quantum effects give an advantage: simulation of nature, optimization/search, secure communication and precision measurement. Most near-term uses are hybrid: classical computers + quantum accelerators (QPUs), like GPUs.

» 1.1 Healthcare and drug discovery

Drugs work through molecular interactions governed by quantum mechanics. Classical computers cannot exactly simulate molecules beyond ~tens of electrons because the state space grows exponentially. A quantum computer represents the molecule's electronic structure naturally.

- Molecular simulation: compute ground-state energies and reaction pathways with high accuracy (algorithms: VQE, Quantum Phase Estimation).
- Protein folding & binding: predicting how a drug candidate docks to a target; quantum-enhanced ML for screening huge chemical spaces.
- Faster, cheaper R&D: average drug costs > US$2 billion and ~10-15 years; better in-silico screening could cut failures.
- Personalized medicine & genomics: optimizing treatment plans, analysing genomic data, radiotherapy planning (optimization).

- *Medical imaging & diagnosis: quantum sensors - MEG with optically pumped magnetometers, NV-diamond MRI, improved MRI contrast.*
- *Collaborations: Roche, Boehringer Ingelheim with Google/Cambridge Quantum; Merck, Pfizer, AstraZeneca pilot projects; Moderna with IBM (mRNA).*

» 1.2 Material science

- *Design of new materials by simulation: high-temperature superconductors, better batteries (lithium-ion/solid-state electrolytes), solar-cell materials, light-weight alloys.*
- *Catalysis: understanding the nitrogenase enzyme (FeMoco) could enable energy-efficient fertilizer production (Haber-Bosch uses ~1-2 % of world energy); carbon capture catalysts.*
- *Condensed-matter models (Hubbard model, spin liquids) - analog quantum simulators with cold atoms already beat classical in some regimes.*
- *Quantum materials for quantum devices: superconductors, diamond, silicon-28, topological materials - a virtuous loop.*
- *Companies: BASF, Daimler/Mercedes (batteries), Boeing/Airbus (materials), Toyota.*

> ★ *Why quantum helps chemistry*
> *Electron correlation is intrinsically quantum. N spin-orbitals require 2^N amplitudes classically; N qubits store them directly. Rough estimate: ~100-200 logical qubits for useful small molecules, ~4000 logical qubits for FeMoco-scale problems.*

» 1.3 Logistics and optimization

- *Many business problems = choose the best among astronomically many options (travelling salesman, vehicle routing, scheduling, portfolio selection, supply-chain, flight gates, power-grid balancing).*
- *Quantum approaches: QAOA, quantum annealing (D-Wave), Grover-based search, hybrid solvers.*
- *Pilot cases: Volkswagen traffic-flow optimization (Lisbon), DHL/Airbus route & loading, Port of Los Angeles scheduling, Goldman Sachs/JPMorgan option pricing and risk (Monte-Carlo with quadratic speed-up), energy-grid optimization.*
- *Honest status: advantage not yet proven for real-sized problems; strong classical heuristics compete. Value may appear when error-corrected machines arrive.*

» 1.4 Quantum sensing and precision timing

Quantum sensing uses quantum states (superposition/entanglement) that are extremely sensitive to external fields, giving precision beyond classical limits (standard quantum limit → Heisenberg limit). It is the most mature quantum technology.

Sensor	Principle	Applications
Atomic clocks (optical lattice)	Frequency of atomic transition	GPS/GNSS, telecom sync, finance time-stamping, tests of relativity; error ~1 s in billions of years
NV-centre magnetometers	Spin of nitrogen-vacancy defect in diamond	Nanoscale MRI, brain & heart signals, battery inspection, mineral exploration
SQUIDs / OPMs	Superconducting loops / atomic vapours	Magnetoencephalography (MEG), geophysics
Atom interferometers	Matter-wave interference of cold atoms	Gravimeters & gradiometers: underground mapping, oil/water, inertial navigation without GPS
Quantum LiDAR / radar / imaging	Entangled or single photons	Stealth detection, imaging through fog, low-light bio-imaging
Quantum gravimeters	Cold-atom free-fall	Tunnel detection, volcano monitoring, resource exploration

Precision timing

The SI second is defined by caesium-133 (9,192,631,770 oscillations). Optical clocks (Sr, Yb, Al^+) are 100× more stable: they would neither gain nor lose a second in the age of the universe; they enable chronometric geodesy (measure height differences of cm by gravitational redshift).

2. Industrial Case Studies

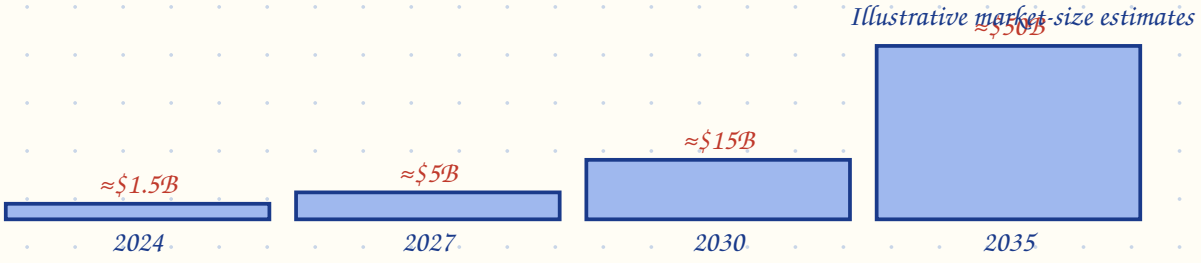


Fig 2: Rough growth of quantum market forecasts (illustrative)

Company	Approach / technology	Highlights
IBM	Superconducting transmon qubits; Qiskit; IBM Quantum Network; Quantum System Two (modular)	Eagle 127q (2021), Osprey 433q (2022), Condor 1121q (2023); Heron 133/156q higher-quality chips; roadmap to error-corrected 'Starling' (~200 logical qubits, 2029); qLDPC codes; cloud access to 100+ devices; Quantum-safe cryptography contributions; partnerships with IITs/IISc & Indian cloud hubs.

Company	Approach / technology	Highlights
Google Quantum AI	Superconducting; Cirq; Sycamore → Willow	2019 'quantum supremacy' (200 s vs 10,000 years claim); 2023 logical qubit experiments; Dec 2024 Willow 105q: exponential suppression of errors with larger surface code - 'below threshold'; random-circuit sampling in 5 minutes; 2025 verifiable 'Quantum Echoes' advantage claim; goal: useful error-corrected computer by ~2029-2030.
Microsoft	Topological qubits (Majorana); Azure Quantum cloud; Q# language	Partners with Quantinuum, Atom Computing for logical-qubit demos (2024); Majorana 1 chip announcement (2025, debated by experts); resource-estimator tool; quantum-safe crypto; Azure Quantum Elements (chemistry/AI).
PsiQuantum	Photonic, fusion-based, manufactured in GlobalFoundries silicon-photonics fab	Raised > US$1 billion (2025); building utility-scale million-qubit machines in Brisbane (Australia) & Chicago; pursuing cryogenic (~4 K) photonic modules.
Others	IonQ & Quantinuum (ions), Rigetti, D-Wave (annealing, claim of advantage on material simulation 2025), QuEra/Pasqal/Atom Computing (neutral atoms), Xanadu, Intel, Amazon (AWS Ocelot cat-qubit chip), NVIDIA (CUDA-Q hybrid)	India: QpiAI (25-qubit 'Indus' superconducting computer), TCS, L&T, Tata, Mahindra, BosonQ Psi, Pixxel/ISRO collaborations.

Exam tip / Remember

For exam: pick two companies, write 5 lines each - platform, key milestone (qubits/year), software (Qiskit/Cirq/Q#), roadmap. Also state that milestone claims are debated and evolving.

3. Ethical, Societal and Policy Considerations

» Ethical & societal issues

- Security & privacy: Shor's algorithm threatens present encryption; 'harvest now, decrypt later' surveillance risk; need quantum-safe migration.
- Digital divide / quantum divide: only wealthy nations and companies may access quantum power → inequality; open cloud access and education help.
- Dual use: military applications (code-breaking, navigation, stealth detection, sensing of submarines) → arms-race and export controls.

- *Economic disruption & jobs: new industries but possible disruption in finance, cybersecurity, pharma.*
- *Responsible innovation: anticipate misuse early (lesson from AI); inclusive governance; transparency vs hype ('quantum-washing', exaggerated claims).*
- *Environmental aspects: cryogenics use energy and helium-3; but quantum may reduce the energy needed for some computations and chemistry (fertilizer, batteries).*
- *Ethics of access & ownership: patents, open-source standards, data sovereignty.*

» Policy landscape

Policy area	Examples
National strategies	India NQM (2023), US NQI Act (2018) & CHIPS/Quantum acts, EU Quantum Flagship & Strategy (2023), UK 10-year plan, China 14th Five-Year Plan, Japan, Korea, Australia, Canada
Export controls	US/EU/UK controls on advanced quantum computers & components (2024), Wassenaar-style arrangements
Quantum-safe migration	NIST PQC standards (FIPS 203/204/205, Aug 2024); US NSM-10 / 2035 deadline; India's NCIIPC/DST advisories; CNSA 2.0
Standards	ISO/IEC JTC 3, IEEE P7130, ETSI QKD specs, ITU-T; BIS India working groups
Workforce	Fellowships, quantum education acts, industry-academia hubs
Ethics frameworks	UNESCO/OECD/WEF 'Quantum for Society' & responsible quantum technology principles

4. Challenges to Adoption

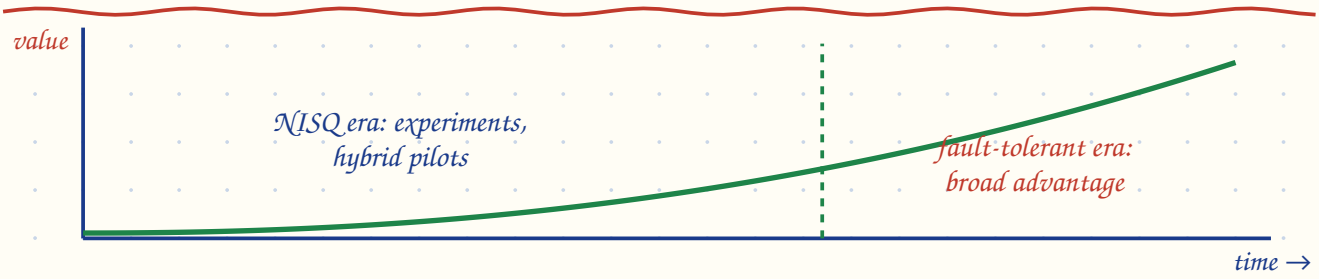


Fig 3: Value expected to rise sharply only after fault tolerance

Challenge	Details	Way forward
Cost	A superconducting system costs US$10-60 million; dilution fridges, lasers, cleanrooms; running costs and helium	Cloud access (pay-per-use), shared national facilities, modular/room-temperature designs, economies of scale

Challenge	Details	Way forward
Skills / talent	Interdisciplinary: physics + CS + engineering; global shortage (≈3 jobs per qualified candidate)	New degrees, NPTEL/NQM courses, bootcamps, upskilling programmes, industry internships
Standardization	No common benchmarks, gate sets, APIs, security certification for QKD/PQC	ISO/IEC, IEEE, BIS; open-source frameworks; benchmarks (quantum volume, CLOPS, algorithmic qubits)
Technical immaturity	Noise, small scale; unclear 'killer app'	Hybrid algorithms, error mitigation, co-design
Integration	Linking QPUs with HPC & cloud; software stacks	Quantum-HPC centres, middleware, CUDA-Q, Qiskit Runtime
Hype & trust	Unrealistic promises	Transparent roadmaps, independent verification
Supply chain	Helium-3, rare isotopes, specialized lasers, superconducting cables	Domestic manufacturing, material recycling

5. Emerging Careers in Quantum

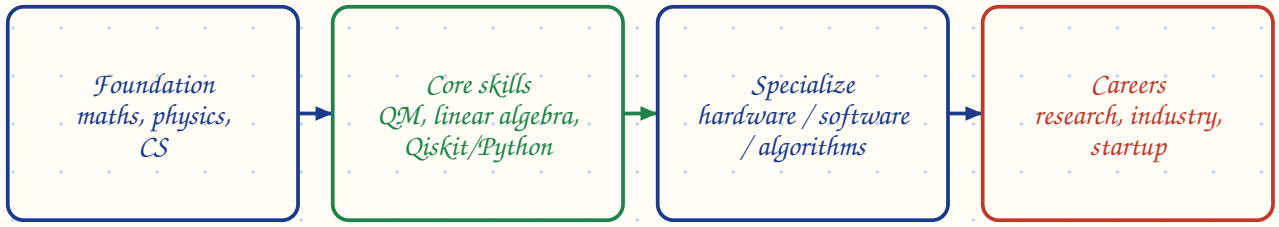


Fig 4: A pathway into quantum technology

Role	What they do	Skills
Quantum research scientist / physicist	Develop theory, qubits, protocols	QM, many-body physics, maths, papers; PhD usual
Quantum hardware / device engineer	Fabricate & test qubits, cryogenics, lasers, photonics chips	Microfabrication, RF/microwave, optics, vacuum, materials
Control & electronics engineer	FPGA, pulse generation, cryo-CMOS, readout chains	Digital/analog design, signal processing
Quantum software developer	Compilers, SDKs, cloud platforms, simulators	Python, C++/Rust, Qiskit/Cirq/PennyLane, software engineering
Quantum algorithm / application scientist	Map industry problems to quantum algorithms; resource estimation	Linear algebra, complexity, domain knowledge (chem, finance)

Role	What they do	Skills
Quantum cybersecurity specialist	PQC migration, QKD deployment, crypto-agility	Cryptography, networks, standards
Quantum network / communication engineer	Photonic links, repeaters, satellites	Optical comms, photonics, protocols
Quantum sensing engineer	Atomic clocks, magnetometers, navigation	Atomic physics, metrology, instrumentation
Product / business / policy roles	Strategy, partnerships, standards, ethics	Tech literacy + business/legal; communication

» Preparation pathway

1. *Foundations:* calculus, linear algebra, probability, classical & modern physics, programming (Python).
2. *Core quantum:* quantum mechanics, quantum information (Nielsen & Chuang; Preskill's notes; Qiskit Textbook; edX/NPTEL/Coursera courses).
3. *Hands-on:* run circuits on IBM Quantum / Amazon Braket / Azure Quantum; Qiskit, Cirq, PennyLane; open-source contributions; hackathons (Qiskit Fall Fest, IBM Quantum Challenge, NQM hackathons).
4. *Specialize:* hardware (cryogenics, photonics), algorithms (chemistry, optimization), software, communication.
5. *Higher studies / experience:* MSc/MTech/PhD at IISc, IITs, TIFR, IISERs, RRI; internships at QpiAI, TCS, IBM, Quantinuum, national labs.
6. *Soft skills:* interdisciplinary teamwork, communication, ethical awareness; certifications (e.g. IBM Quantum Developer Certification).

> *Exam tip / Remember*
> Quantum workforce needs are not only PhDs: technicians, software engineers, cyber specialists and project managers are required. A good mix: strong CS/maths + basic quantum knowledge = employable fast.

6. Educational and Research Landscape; India's Opportunity

» Global research landscape

- Leaders in publications/patents: USA, China, Europe (Germany, Netherlands, UK, France), Japan, Canada, Australia; hubs: MIT, Harvard, Caltech, Delft, Oxford, ETH Zurich, USTC, Waterloo's IQC.

- Public + private investment: > US$40 billion pledged worldwide; venture funding > US$2 billion/yr; 400+ startups.
- Education: MSc/PhD programmes, EU Quantum Technology Master's, Q-12 education initiatives in USA, open online courses, summer schools.

» India's landscape

Institution / initiative	Role
National Quantum Mission (DST, 2023-31, ₹6003 cr)	Four T-Hubs; build 50-1000 qubit computers; 2000 km QKD; sensors, materials; goal: make India a leading nation in QT by 2031
IISc Bengaluru	Quantum computing hub; superconducting & photonic research; Quantum Technology Initiative
IIT Madras (C-QuICT)	Quantum communication hub; QKD, post-quantum crypto
IIT Bombay	Quantum sensing & metrology hub
IIT Delhi	Quantum materials & devices hub
TIFR Mumbai	Superconducting qubits - India's first indigenous 6-qubit/ transmon work; fundamental research
RRI Bengaluru, IISER Pune/Mohali, IIT Kanpur, IIT Kharagpur, IIT Hyderabad, C-DAC	Quantum optics, communication, algorithms, supercomputing & quantum simulators
ISRO, DRDO, DAE	Satellite quantum communication; QKD links; defence applications
Amaravati Quantum Valley (Andhra Pradesh)	State initiative with IBM Quantum System Two; planned to host a 156-qubit machine & ecosystem; startups & skilling
Startups & industry	QpiAI (Bengaluru), BosonQ Psi, QNu Labs (QKD), Quantum Hub; TCS, Infosys, Wipro, L&T, Tata, Mahindra

» India's opportunity

- Talent advantage: one of the world's largest STEM and IT workforces; strong maths/physics tradition (Bose - Bose-Einstein statistics, Raman effect, Sen, Bhabha).
- Software & services strength: Indian IT firms can lead in quantum software, algorithms, cloud and services.
- Cost advantage & scale: lower-cost manufacturing and test infrastructure; large domestic market for secure communication (banking, defence, space).
- Policy push: NQM, ISRO satellite capability, semiconductor mission synergy, startup ecosystem (Startup India).
- Strategic autonomy: indigenous QKD/PQC and sensing reduce dependence on foreign tech.

- *Gaps: fabs & cryogenic supply chain, specialised labs, funding scale (small relative to USA/China), faster industry-academia linkage, retention of talent ('brain-drain').*

> *The big picture*
> *Quantum is a once-in-a-generation platform shift (like semiconductors and AI). Nations that invest early in people, standards and supply chains will shape the rules. India has the talent and momentum - execution and speed are the keys.*

6A. More Application Areas

» Finance

- *Monte-Carlo & option pricing: quantum amplitude estimation gives a quadratic speed-up (error ~1/N instead of 1/√N).*
- *Portfolio optimization, credit scoring, fraud detection, risk analysis (JPMorgan, Goldman Sachs, HSBC, Wells Fargo, Indian banks exploring).*
- *Quantum-safe banking: QKD links between data centres; PQC for UPI/SWIFT-like systems.*

» Energy, climate and environment

- *Better batteries, solar absorbers, fusion-plasma simulation, carbon-capture sorbents.*
- *Smart-grid optimization and renewable forecasting; quantum gravimeters to find water/minerals; climate-model acceleration (long-term).*
- *Fertilizer: cheaper ammonia through catalyst design (FeMoco) → less CO_2.*

» Artificial Intelligence & machine learning

- *Quantum machine learning: quantum kernels, variational classifiers, generative models - speculative, benefits not yet established.*
- *AI helps quantum: decoders for error correction, calibration, circuit compilation, discovering new algorithms.*
- *Hybrid pipelines: GPUs + QPUs (NVIDIA DGX Quantum, CUDA-Q).*

» Aerospace, defence & navigation

- *GPS-free inertial navigation using cold-atom accelerometers - for submarines, aircraft, autonomous vehicles.*
- *Secure satellite links, quantum radar/LiDAR, detection of hidden objects by gravity/magnetic anomalies.*
- *Concerns: dual-use & escalation; international norms needed.*

Domain	Technology	Timeframe (approx.)
Timing, magnetometry, gravimetry	Quantum sensing	Now - 5 years
QKD, QRNG, PQC migration	Quantum communication / security	Now - 10 years
Chemistry & materials small-scale	Quantum computing (NISQ + early FT)	5 - 10 years
Optimization, finance, large simulations	Fault-tolerant computing	10 - 20 years
Quantum internet, distributed QC	Networks + memory + repeaters	15 - 25 years

6B. Glossary for Unit 5

Term	Meaning
Quantum advantage	A quantum device performs a useful task better (faster/cheaper/more accurately) than any classical method
Quantum supremacy	Quantum device solves some (possibly useless) problem infeasible for classical computers
NISQ	Noisy Intermediate-Scale Quantum devices (50-1000 imperfect qubits)
Logical qubit	Error-protected qubit made from many physical qubits
Quantum volume	IBM's single-number benchmark of overall quality & size of a device
Hybrid algorithm	Algorithm splitting work between classical and quantum processors (VQE, QAOA)
PQC	Post-quantum cryptography: classical algorithms resistant to quantum attack
Crypto-agility	Ability to swap cryptographic algorithms quickly
Dual-use	Technology with civilian and military uses
T-Hub	Thematic Hub of the Indian National Quantum Mission
QaaS	Quantum computing as a Service via cloud

6C. Quick Timeline of Quantum Milestones

Year	Milestone
1900-1927	Planck, Einstein, Bohr, de Broglie, Schrödinger, Heisenberg - quantum mechanics born
1947-60	Transistor and laser - first quantum revolution
1981-85	Feynman & Deutsch propose quantum computers
1984, 1991	BB84 and E91 quantum cryptography
1994-96	Shor's and Grover's algorithms; Shor's error-correcting code (1995)
1998-2007	First few-qubit NMR, ion-trap and superconducting qubits
2016-19	Micius satellite; IBM Q Experience; Google 'supremacy' (2019)
2022	Nobel Prize in Physics - Aspect, Clauser, Zeilinger (entanglement experiments)
2023	India's National Quantum Mission; IBM Condor 1121 qubits
2024	NIST PQC standards; Google Willow below-threshold error correction
2025	UN International Year of Quantum Science & Technology; new logical-qubit and cat-qubit chips

7. Case-Study Style Mini Answers

Q. How can quantum computing speed up drug discovery?

(1) Simulate electronic structure of target molecules accurately (VQE, QPE) → better binding-energy estimates. (2) Reduce wet-lab trials by in-silico screening. (3) Hybrid quantum-classical ML for candidate selection. Present status: small molecules (H_2, LiH, BeH_2) done on hardware; practical benefit expected with error-corrected machines.

Q. Why are quantum sensors the nearest-term technology?

They need only a few well-controlled quantum systems (not millions of qubits), operate on short timescales, and gain from sensitivity not error-correction. Already commercial: atomic clocks, magnetometers, gravimeters.

Q. Discuss the challenges to adoption.

Cost (hardware, cryogenics), skills shortage, lack of standards, technical immaturity, integration with HPC, hype, supply chain. Mitigation: cloud access, education, standards bodies, hybrid algorithms, public funding, international collaboration.

Q. Is quantum computing a threat to current security?
Yes in the future: Shor can break RSA/ECC. Remedy: post-quantum cryptography (NIST 2024), crypto-agility, QKD for specific links; start migration now due to 'harvest now, decrypt later'.

8. Quick Revision & Likely Questions

1. *List the application domains of quantum technologies with examples.*
2. *How can quantum computers help in drug discovery and material science?*
3. *Explain how quantum sensing & atomic clocks work and their uses.*
4. *Discuss industrial efforts of IBM, Google, Microsoft and PsiQuantum.*
5. *Explain ethical, societal and policy issues related to quantum technology.*
6. *What are the barriers to adoption: cost, skills, standardization?*
7. *Describe career roles in quantum technology and the preparation pathway.*
8. *Explain India's opportunity and role in the global quantum race.*

★ *Final takeaway*
Quantum technologies are moving from labs to markets: sensing today, communication soon, computing next. Success depends on technology, talent, trust and policy together.